

Neural Clustering

based Visual Representation Learning

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Zhejiang University | ETH Zürich | CVPR 2024

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The Problem with Existing Approaches

All existing visual feature extractors — ConvNets, ViTs, and MLPs — share the same fundamental limitation: they represent images as rigid rectangular grids.

ConvNets

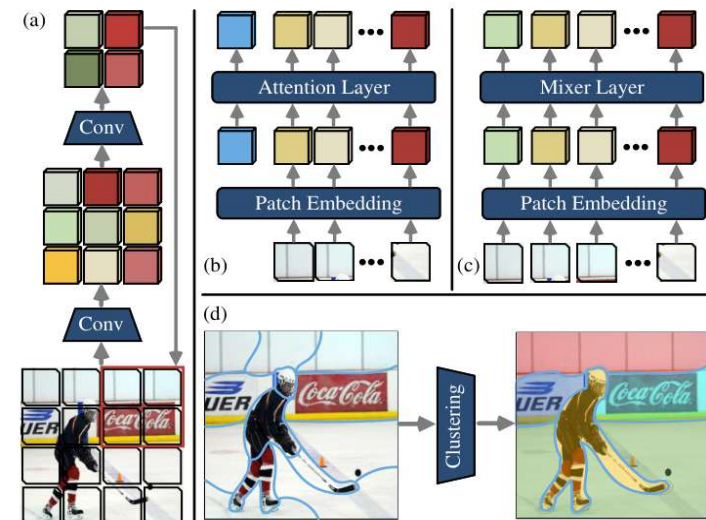
Sliding window with fixed kernels over rectangular regions

ViTs

Divides image into fixed non-overlapping square patches

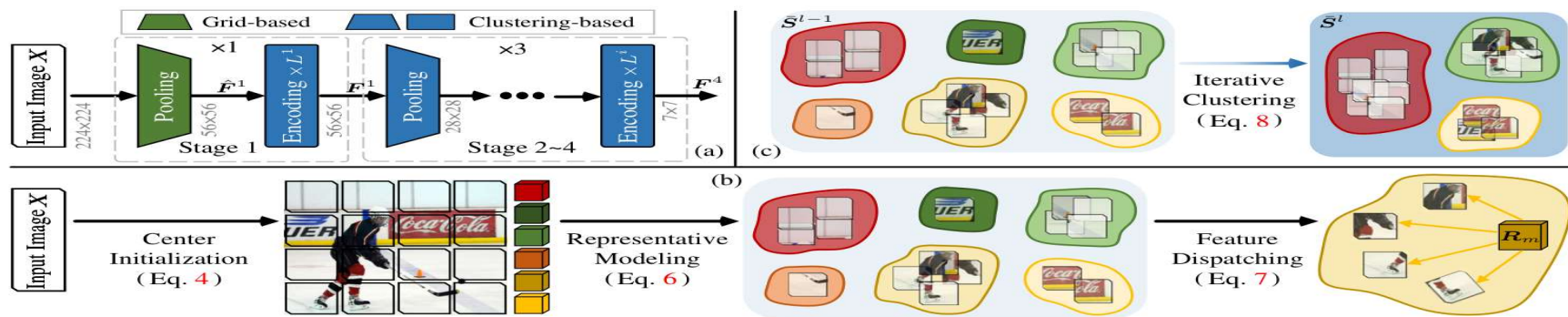
MLPs

Token mixing over predetermined grid-shaped patch embeddings



FEC: Feature Extraction with Clustering

Instead of rigid grids, FEC views feature extraction as selecting representatives from data — automatically capturing the underlying distribution.



1. small patches

starts by dividing image into small patches using standard convolution (sliding window operation)

2. Cluster based feature encoding

Forms a cluster representative and does feature dispatching

3. Cluster based feature pooling

Uses the aggregated cluster representation to form a low dimensional feature map.

Why FEC? — Three Key Advantages



Ad-hoc Interpretability

Unlike Grad-CAM and other post-hoc methods that require complex post-processing, FEC's forward process is fully transparent. Cluster assignments at every layer can be viewed and inspected directly by humans



Captures True Data Distribution

Automatically discover underlying data distribution , revealing relationship between pixels and capturing semantic structure that grid based may miss.



Versatile & Efficient

FEC-Small achieves 72.7% ImageNet Top-1 accuracy with only 5.5M parameters. FEC is valuable in safety critical application and provide insights into how model process visual data.

How FEC Works — The Pipeline

Clustering

1

The model groups similar pixels together based on feature similarity (like color, texture etc) and each group forms a cluster.

Representative Modeling

2

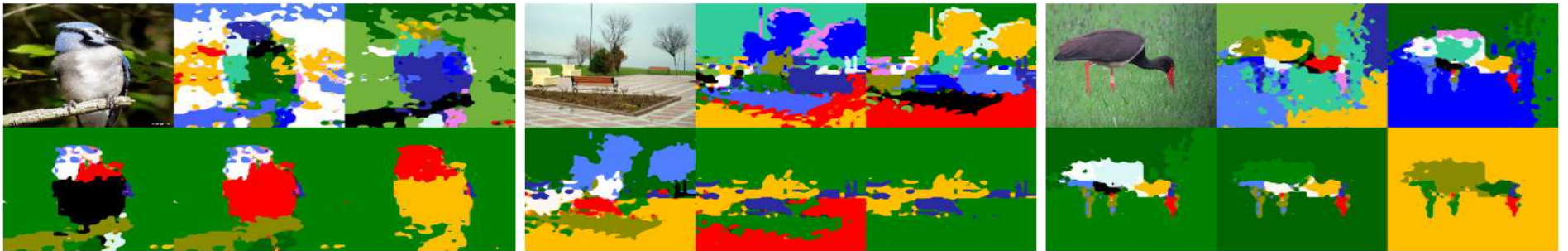
For each cluster , a representative (center) is created and this representative summarizes all pixels in that cluster

Feature Dispatching

3

The representative send information back to the pixels. The pixel features are updated based on their cluster and the pixels in same cluster become more consistent.

Seeing Inside FEC — Cluster Visualizations



Insight 1

As layers deepen, small pixels groups combine into bigger, meaningful parts – like how humans see objects step by step.

Insight 2

Final cluster clearly represent objects like a bird or its parts , even though the model was not explicitly told to do so.

Performance Results Across Tasks

72.7%

ImageNet Top-1

FEC-Small (5.5M params)

78.1%

ImageNet Top-1

FEC-Base (14.4M params)

81.2%

ImageNet Top-1

FEC-Large (28.3M params)

+2.9%

vs ResNet18

with 50% fewer parameters

Task	Dataset	FEC-Small	FEC-Base	FEC-Large
Classification	ImageNet	72.7% (5.5M)	78.1% (14.4M)	81.2% (28.3M)
Semantic Seg.	ADE20K	35.3% (9.1M)	37.7% (18.0M)	40.5% (31.9M)
Object Detection	MS COCO	35.6% (24.3M)	37.9% (33.1M)	39.9% (47.1M)

Limitations & Future Directions

Limitations

Simple clustering may not handle complex scenes well , where objects are detailed or overlapping

Fixed cluster count must be predefined (parametric), introducing subjectivity

A fixed number of clusters may not fit all images , since each image has a different structure

Future Work

Nonparametric clustering (e.g., Dirichlet Process) — auto-discover cluster count

Integrate with set-prediction architectures like DETR for end-to-end detection

Advanced algorithms for complex data structures with better accuracy

FEC marks a solid first step toward fully interpretable, clustering-based visual feature extraction — with much more to explore.

Summary

What FEC brings to machine vision



Replaces the rigid grid paradigm with dynamic, data-driven clustering — aligning feature extraction with how images actually look.



Fully transparent forward process: cluster assignments at every layer are human-readable without any post-processing.



Competitive performance: FEC-Small (5.5M) beats ResNet18 (12M) by 2.9% on ImageNet Top-1.



Drop-in compatibility: residual connections allow FEC to work with detection (YOLO, Faster-RCNN) and segmentation pipelines unchanged.

Implementation of the research paper

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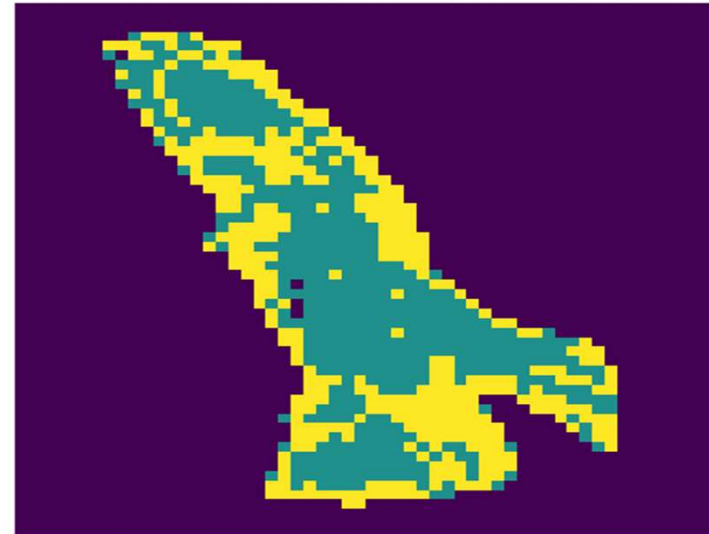
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Part-1 FEC-style Segmentation (Clustering based region)

Original Image



Clustered Segments (FEC Style)



Insight 1

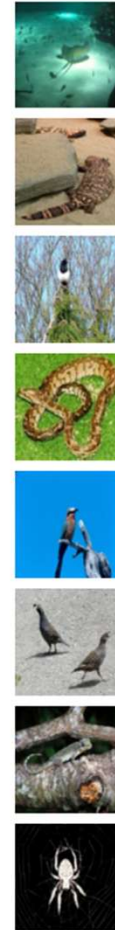
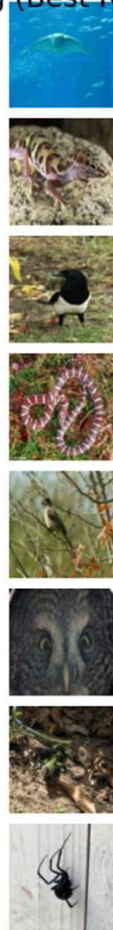
similar pixels are grouped based on visual features(color, texture)

Insight 2

Clear separation of foreground (object) and background

Part-2 Feature clustering

Final Clustering (Best Representatives)



Results

Silhouette score = 0.1009

Even with low score, visual clusters show meaningful grouping

Some clusters are semantically meaningful like spiders grouped together , birds grouped together.

Some overlap exists due to complex data

Conclusion:



The original FEC model involves clustering inside a deep neural network and requires large-scale training on datasets like ImageNet. It also needs high computational resources and a complex architecture.



Due to these constraints, we did not implement the full FEC model. Instead, our goal was to understand and demonstrate the core idea behind FEC.



We implemented a simplified version of FEC where we use pretrained CNN features and apply clustering to group images and image regions.

Thank You
